

Corrupted file

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

This file seems broken... or is it? Maybe a couple of bytes could make all the difference. Can you figure out how to bring it back to life?

Hints

- Try checking the file's header.
 - JPEG
 - Tools like xxd or hexdump can help you inspect and edit file bytes.
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Tools Used

- **file:** Utility to identify file types
 - **strings:** Command-line tool to extract readable ASCII sequences from files
 - **grep:** Text search utility for filtering output
 - **exiftool:** Metadata inspection tool
 - **xxd:** Hexadecimal viewer and editor
 - **hexedit:** Interactive hex editor for modifying binary files
 - **vim:** Text editor for editing hex dumps
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Complete Solution

Step 1: Download and Prepare

Download the file named `file` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type and general information:

```
file file
```

Expected output:

```
file: data
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings file | grep -i "picoCTF"
```

 *If no results appear, proceed to the next step (decoding).*

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata:

```
exiftool file
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : file
Directory                    : .
File Size                    : 8.8 kB
File Modification Date/Time   : 2025:11:07 21:17:32+02:00
File Access Date/Time        : 2026:05:05 11:59:35+02:00
File Inode Change Date/Time   : 2026:05:05 11:59:30+02:00
File Permissions              : -rw-r--r--
Error                        : Unknown file type
```

The metadata itself is not useful here, but the `Error : Unknown file type` message indicates a malformed header. Use `xxd` to inspect the first bytes of the file.

```
xxd file | head -20
```

```

00000000: 5c78 ffe0 0010 4a46 4946 0001 0100 0001 \x....JFIF.....
00000010: 0001 0000 ffdb 0043 0008 0606 0706 0508 .....C.....
00000020: 0707 0709 0908 0a0c 140d 0c0b 0b0c 1912 .....
00000030: 130f 141d 1a1f 1e1d 1a1c 1c20 242e 2720 ..... $. '
00000040: 222c 231c 1c28 3729 2c30 3134 3434 1f27 ",#...(7),01444.'
00000050: 393d 3832 3c2e 3334 32ff db00 4301 0909 9=82<.342...C...
00000060: 090c 0b0c 180d 0d18 3221 1c21 3232 3232 .....2!..!2222
00000070: 3232 3232 3232 3232 3232 3232 3232 3232 2222222222222222
00000080: 3232 3232 3232 3232 3232 3232 3232 3232 2222222222222222
00000090: 3232 3232 3232 3232 3232 3232 3232 ffc0 22222222222222..
000000a0: 0011 0801 f403 2003 0122 0002 1101 0311 ..... ..".....
000000b0: 01ff c400 1f00 0001 0501 0101 0101 0100 .....
000000c0: 0000 0000 0000 0001 0203 0405 0607 0809 .....
000000d0: 0a0b ffc4 00b5 1000 0201 0303 0204 0305 .....
000000e0: 0504 0400 0001 7d01 0203 0004 1105 1221 .....}.....!
000000f0: 3141 0613 5161 0722 7114 3281 91a1 0823 1A..Qa."q.2....#
00000100: 42b1 c115 52d1 f024 3362 7282 090a 1617 B...R..$3br.....
00000110: 1819 1a25 2627 2829 2a34 3536 3738 393a ...%&'()*456789:
00000120: 4344 4546 4748 494a 5354 5556 5758 595a CDEFGHIJSTUVWXYZ
00000130: 6364 6566 6768 696a 7374 7576 7778 797a cdefghijstuvwxyz

```

The first bytes indicate a damaged JPEG signature, so the header needs correction.

```
00000000: 5c78 ffe0 0010 4a46 4946 0001 0100 0001 \x....JFIF.....
```

Image file magic numbers are the specific hexadecimal prefixes at the start of a file that uniquely identify its format. Common signatures include:

- **JPEG (.jpg, .jpeg):** Starts with **FF D8 FF** (specifically **FF D8 FF E0** for JFIF or **FF D8 FF E1** for EXIF metadata).
- **PNG (.png):** Starts with **89 50 4E 47 0D 0A 1A 0A**.
- **GIF (.gif):** Starts with **47 49 46 38** (ASCII "GIF8", followed by either 37 or 39).
- **BMP (.bmp):** Starts with **42 4D** (ASCII "BM").
- **TIFF (.tif):** Starts with **4D 4D 00 2A** (big-endian) or **49 49 2A 00** (little-endian).

These bytes allow software to recognize the file type without relying on the file extension.

Use `hexedit` to correct the header:

```
hexedit file
```

The original bytes are:

```
5C 78 FF E0
```

The corrected JPEG header is:

```
FF D8 FF E0
```

You can also convert the file into editable text with `xxd` , then fix the header in a text editor:

```
xxd file > hex.txt
```

```
vim hex.txt
```

After editing the header, convert the text back to binary:

```
xxd -r hex.txt img.jpeg
```

Open the repaired image to reveal the flag.



```
picoCTF{r3st0r1ng_th3_by73s_efd8c6c0}
```

Final Result

```
picoCTF{<REDACTED>}
```

Key concepts

- **File signature:** The first bytes of a file indicate its real type, regardless of its extension.
- **Corrupted headers:** A modification or corruption of the signature can make a file unrecognized.
- **Hexadecimal:** `xxd` and `hexedit` allow you to inspect and correct bytes directly in hexadecimal.
- **File repair:** Modifying only the header may be enough to restore a valid file.

You can also use this link to inspect and modify bytes in your browser:

<https://hexed.it/>